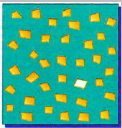
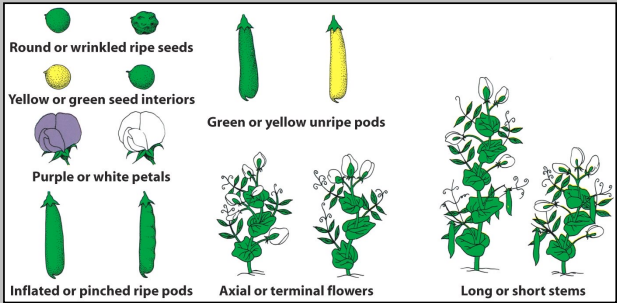


Interacções alélicas

Dominância e recessividade



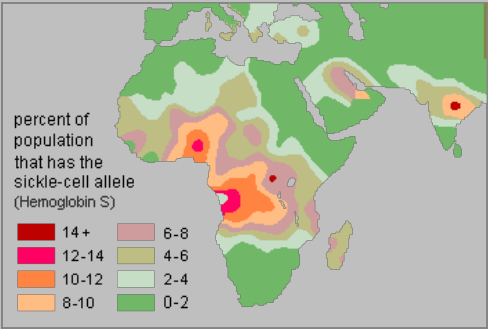
Dominância incompleta



Co-dominância

Phenotypes at Different Levels of Analysis	Normal <i>Hbβ^A Hbβ^A</i>	Carrier <i>Hbβ^A Hbβ^S</i>	Diseased <i>Hbβ^S Hbβ^S</i>
Red blood cell concentration at high altitudes	Normal 	Lower 	Very low, anemia
Red blood cell shape at high altitudes	Normal 	Sickled cells present 	Severe sickling

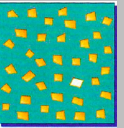
Sobre-Dominância



Letalidade

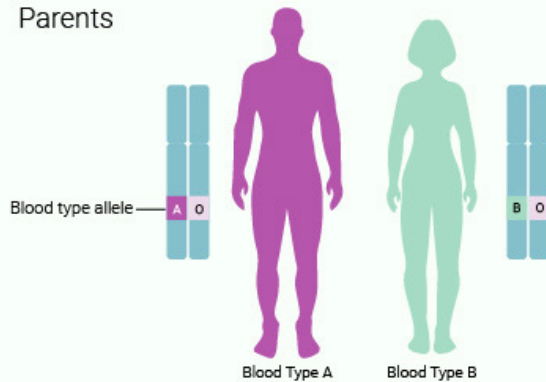


Co-dominância

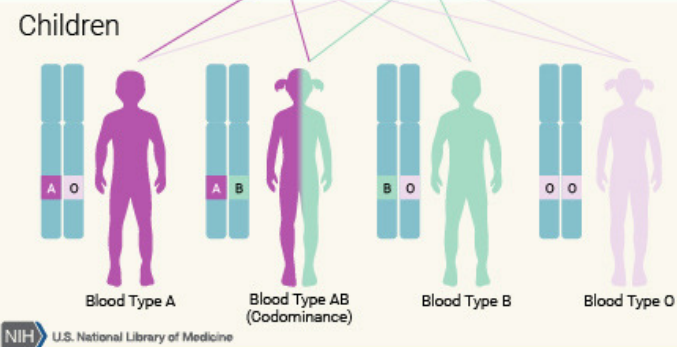


Codominance - example Blood Type

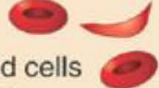
Parents



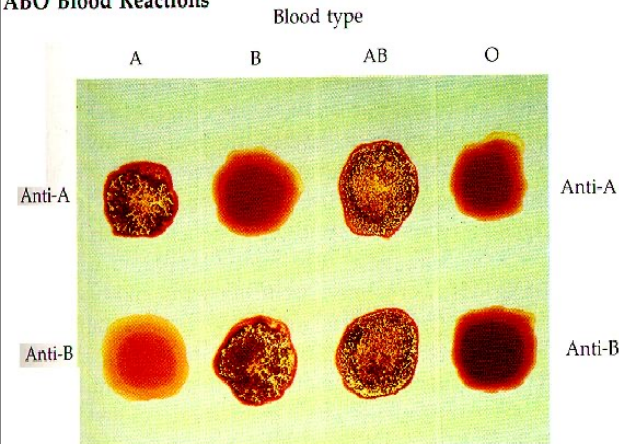
Children



NIH U.S. National Library of Medicine

Phenotypes at Different Levels of Analysis	Normal $Hb\beta^A Hb\beta^A$	Carrier $Hb\beta^A Hb\beta^S$	Diseased $Hb\beta^S Hb\beta^S$
Red blood cell shape at high altitudes	Normal 	Sickled cells present 	Severe sickling 

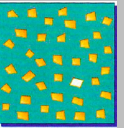
ABO Blood Reactions



3 alelos I^A , I^B e i

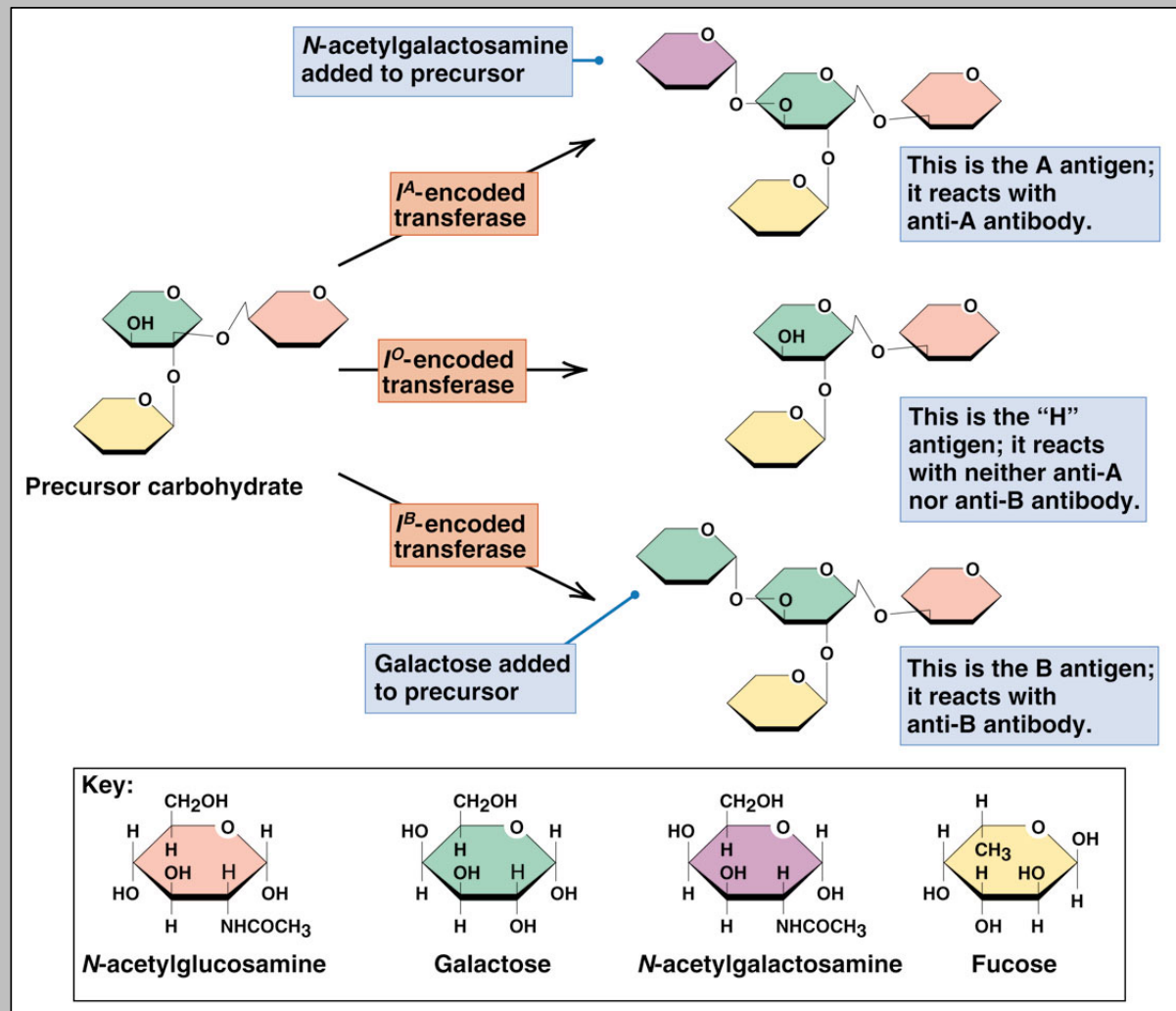
Os heterozigotos expressam o fenótipo de ambos os homozigotos

Alelos múltiplos

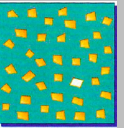


Grupos sanguíneos ABO

Acção do locus $I^A/I^B/i$
(codifica uma transferase)
sobre a Estrutura H



Alelos múltiplos



Alelos múltiplos (polialelismo)

Um gene pode ter vários alelos com diferentes relações de dominância

Série alélica no locus **white** (**w/+**) de *Drosophila melanogaster*

Símbolo	Nome do Alelo	Cor dos Olhos
w	white	Branco
w ^a	white-apricot	Alperce
w ^{bf}	white-buff	Beje
w ^e	White-eosin	Rosa-amarelado
w ^{sat}	white-satsuma	Rubi-escuro
...	...	...

A intensidade da cor reflecte a eficiência de acção (deposição de pigmentos dos olhos) de cada alelo mutante do locus **white/+** (*uns funcionam a 10%, outros a 20%, ...*)

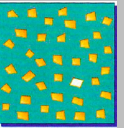
Alelos múltiplos (polialelismo)

Um gene pode ter vários alelos com diferentes relações de dominância

Cor da pelagem dos coelhos

Fenótipo	Genótipo
Castanho-acinzentado	CC ou Cc ^{ch} ou Cc ^h ou Cc
Chinchila	c ^{ch} c ^{ch} ou c ^{ch} c ^h ou c ^{ch} c
Himalaico	c ^h c ^h ou c ^h c
Albino	cc

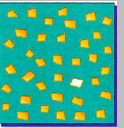
Relações hierárquicas de dominância → C > c^{ch} > c^h > c



Como explicar cores diferentes num mesmo individuo geneticamente homogêneo?

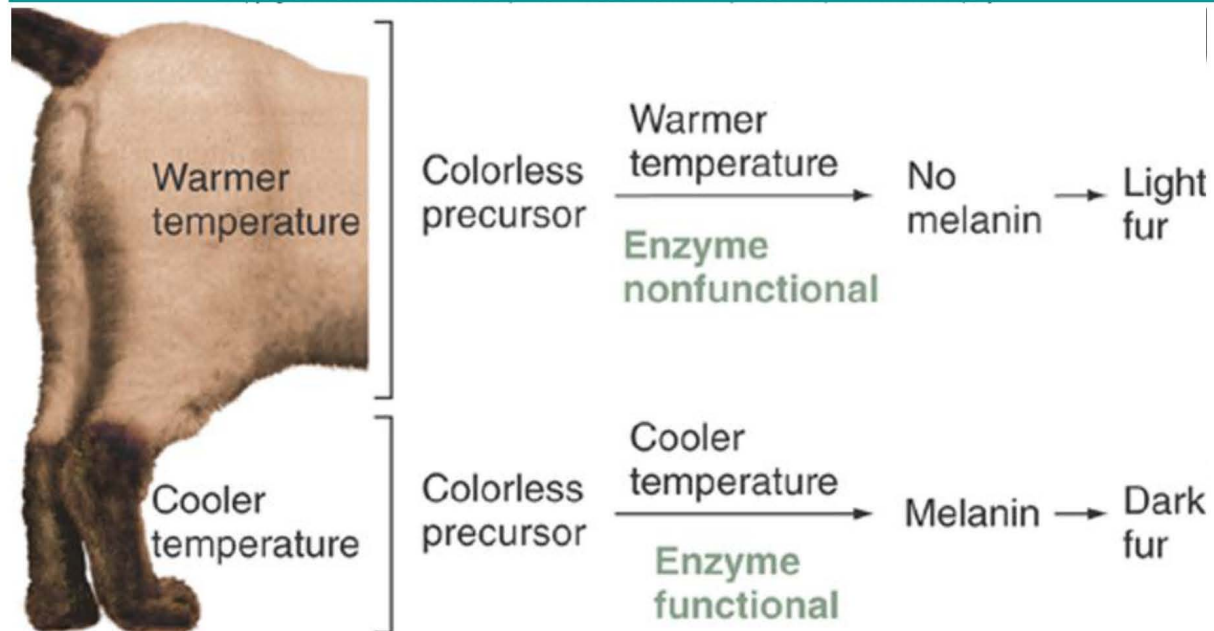


Alelos sensíveis à temperatura (thermo-sensitive)



O alelo himalaia dos coelhos codifica um enzima que só funciona a baixas temperaturas (mutação sensível à temperatura)

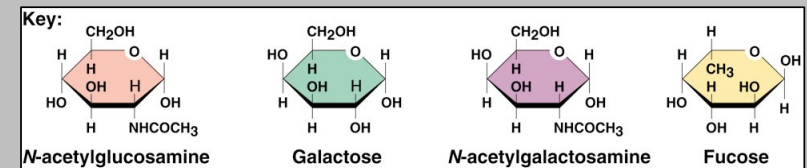
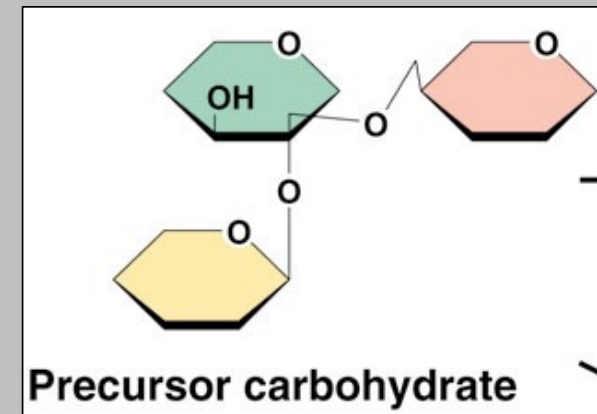
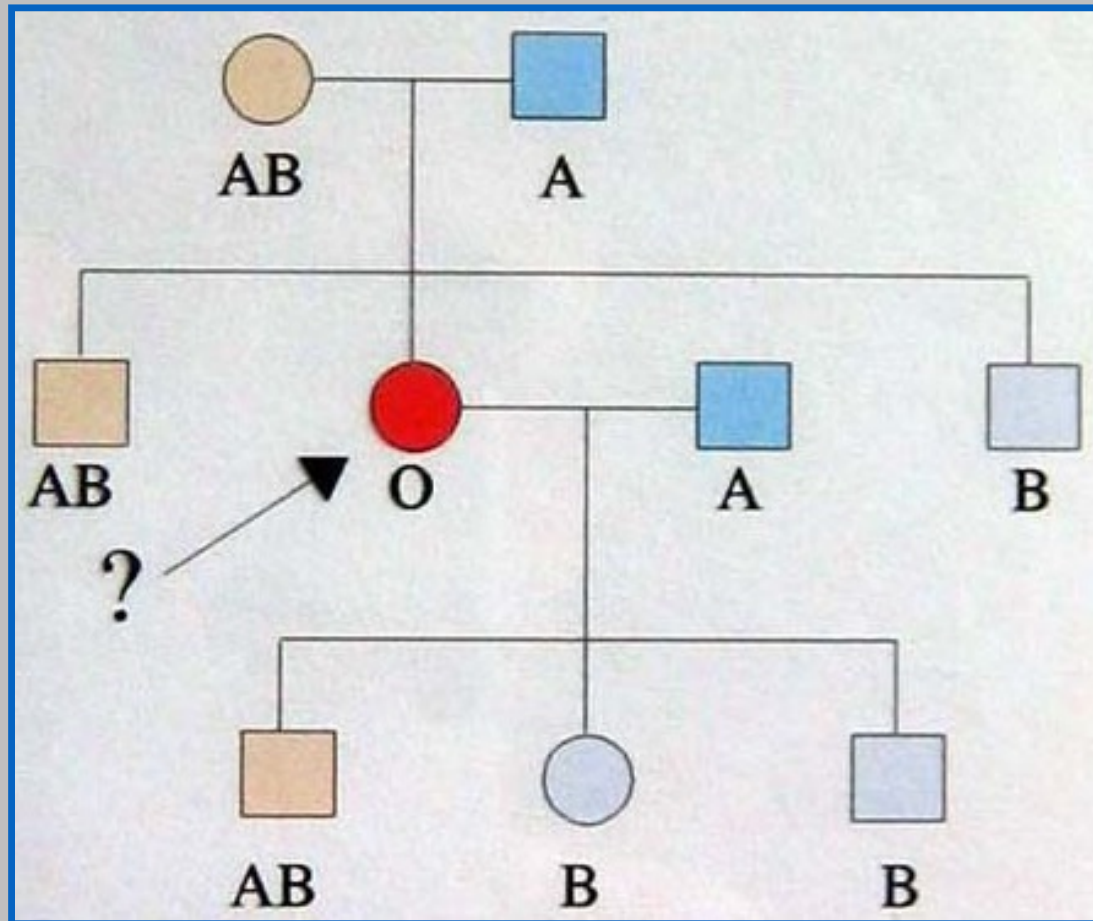
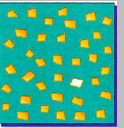
- a pelagem escura ocorre nas zonas mais frias do corpo***
- os gatos siameses resultam dum fenómeno idêntico***





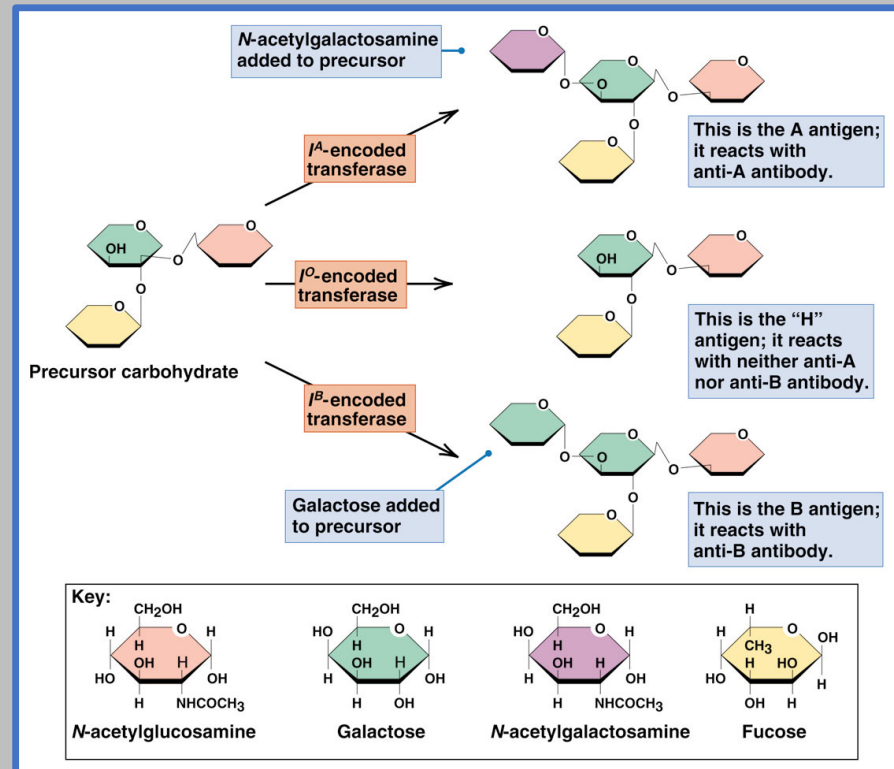
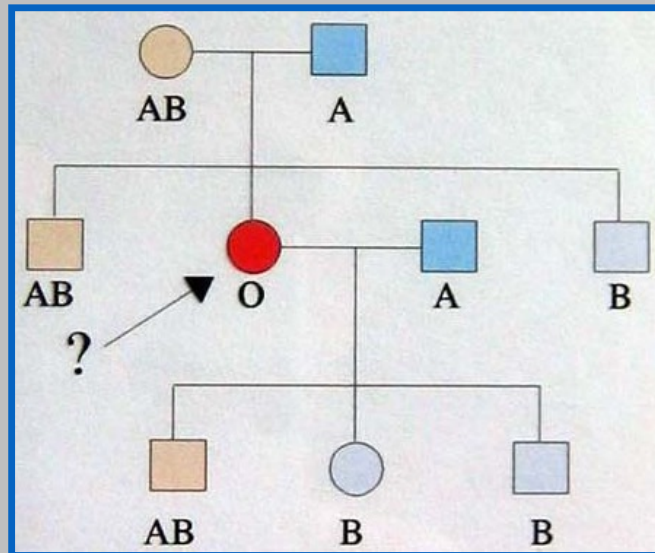
Phenotype = Genetics + Environment + GxE

Alelos múltiplos



Alelos múltiplos

Fenótipo Bombaim = falso tipo sanguíneo O

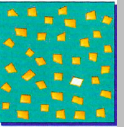


Mutação noutra gene, o que provoca incapacidade de adicionar fucose durante a síntese da estrutura H

Background genético!

Interacções génicas

Interacção Génica



Ocorre quando dois genes contribuem para a mesma característica

Geralmente surgem menos classes fenotípicas → resultado da fusão de classes determinadas por genótipos diferentes

Tipos particulares de Interacção Génica:

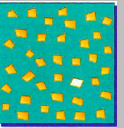
Complementação; supressão; epistasia; redundância

Complementação

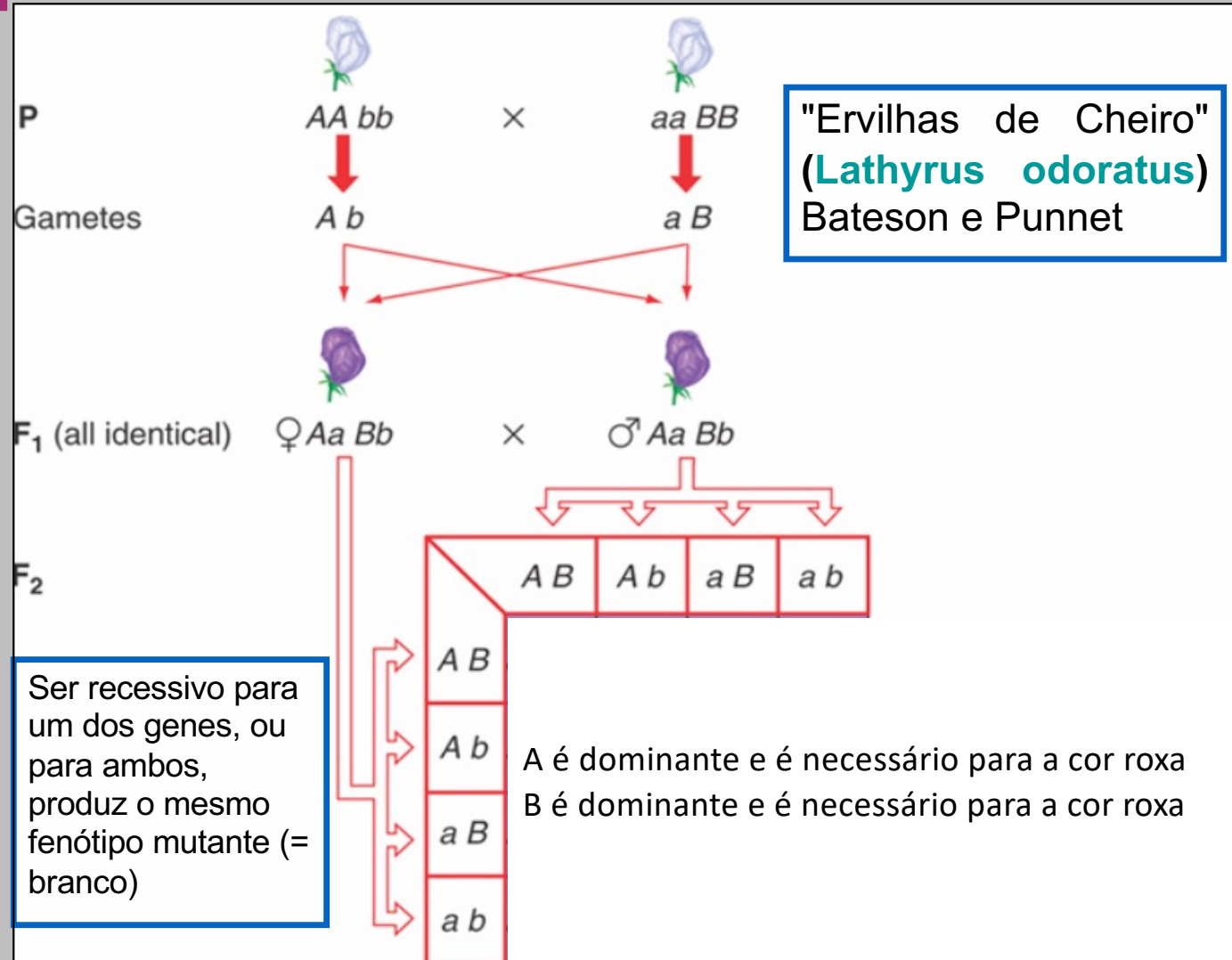
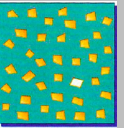
- A produção de fenótipo selvagem pela fusão de dois genomas haplóides numa célula diplóide de mutantes recessivos diferentes.



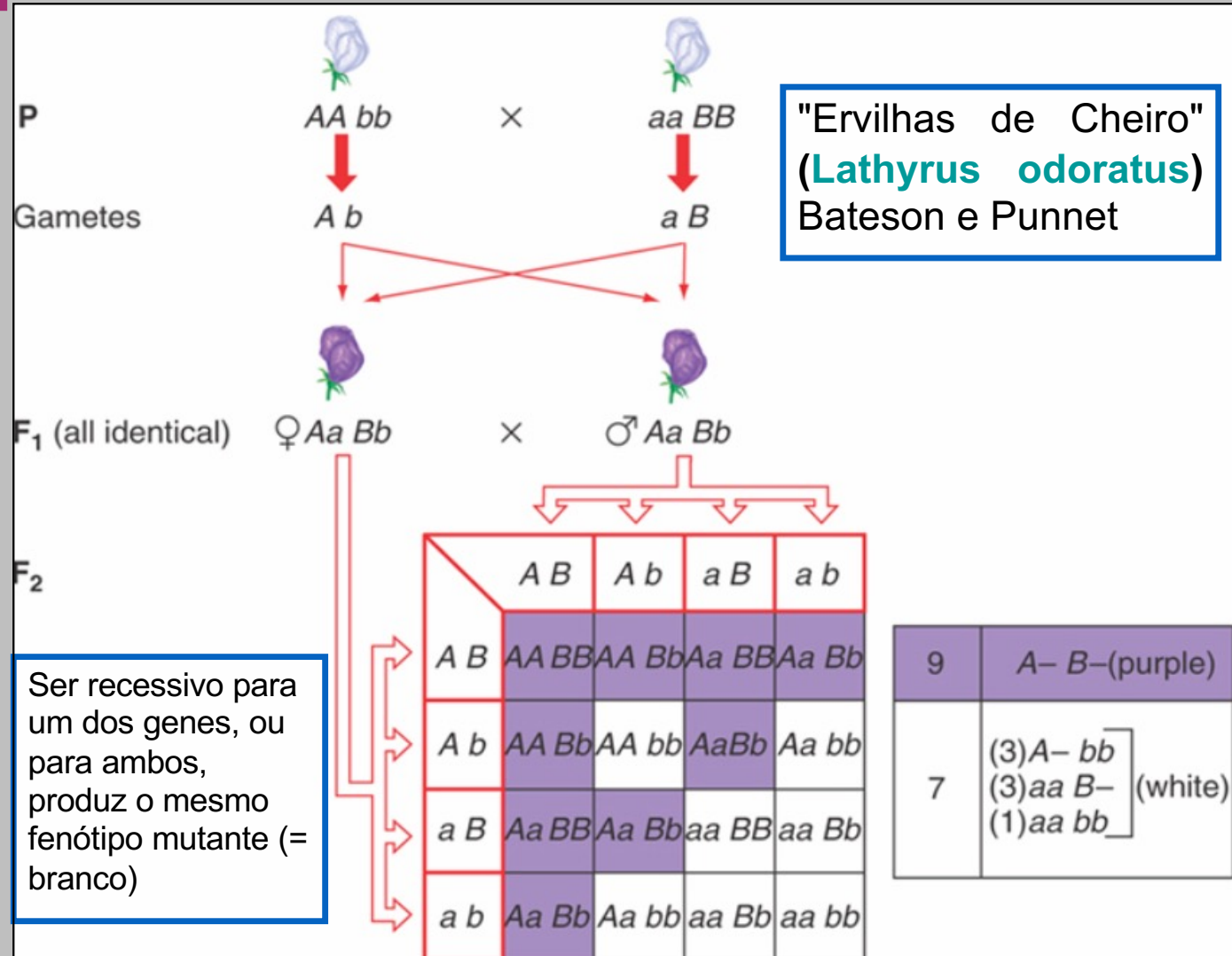
A harebell plant (*Campanula* sp)

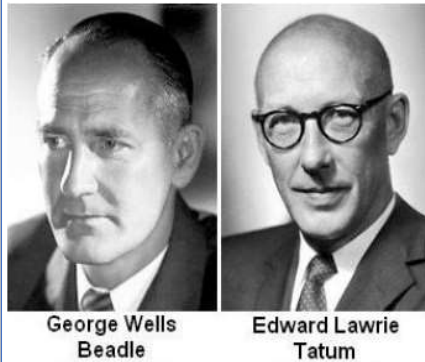


Complementação



Complementação

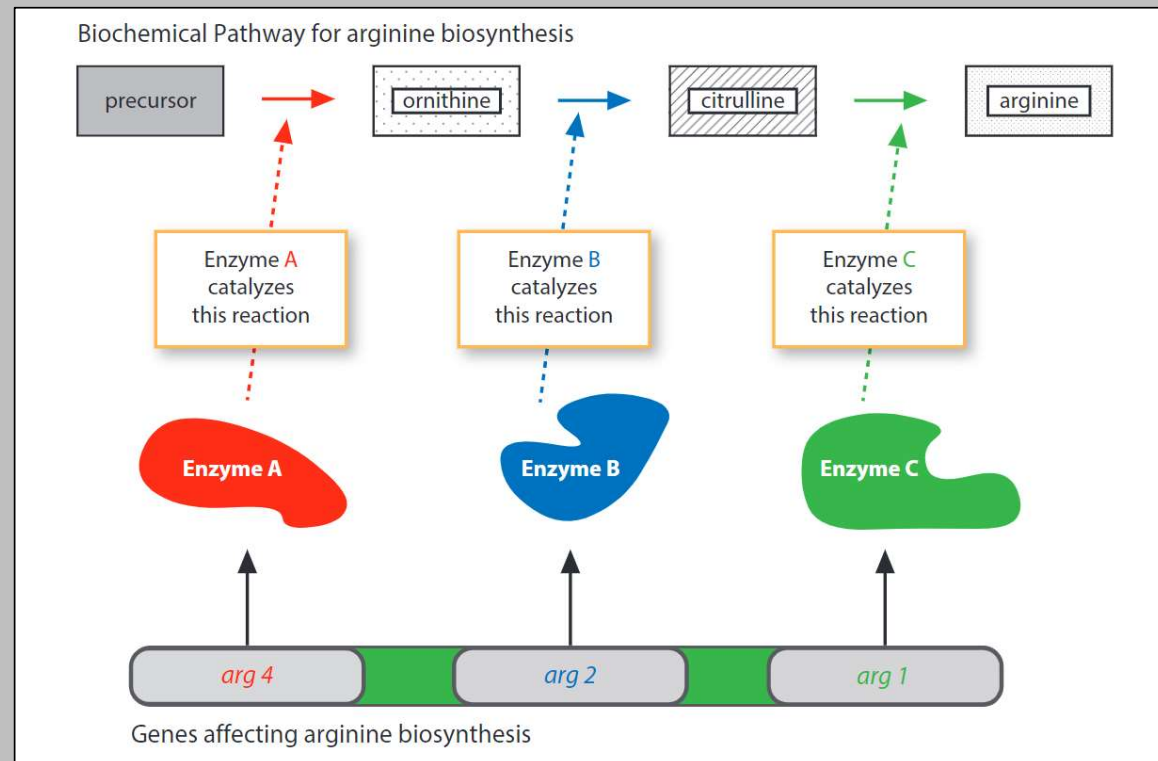




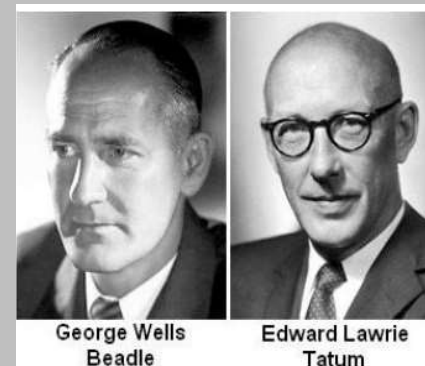
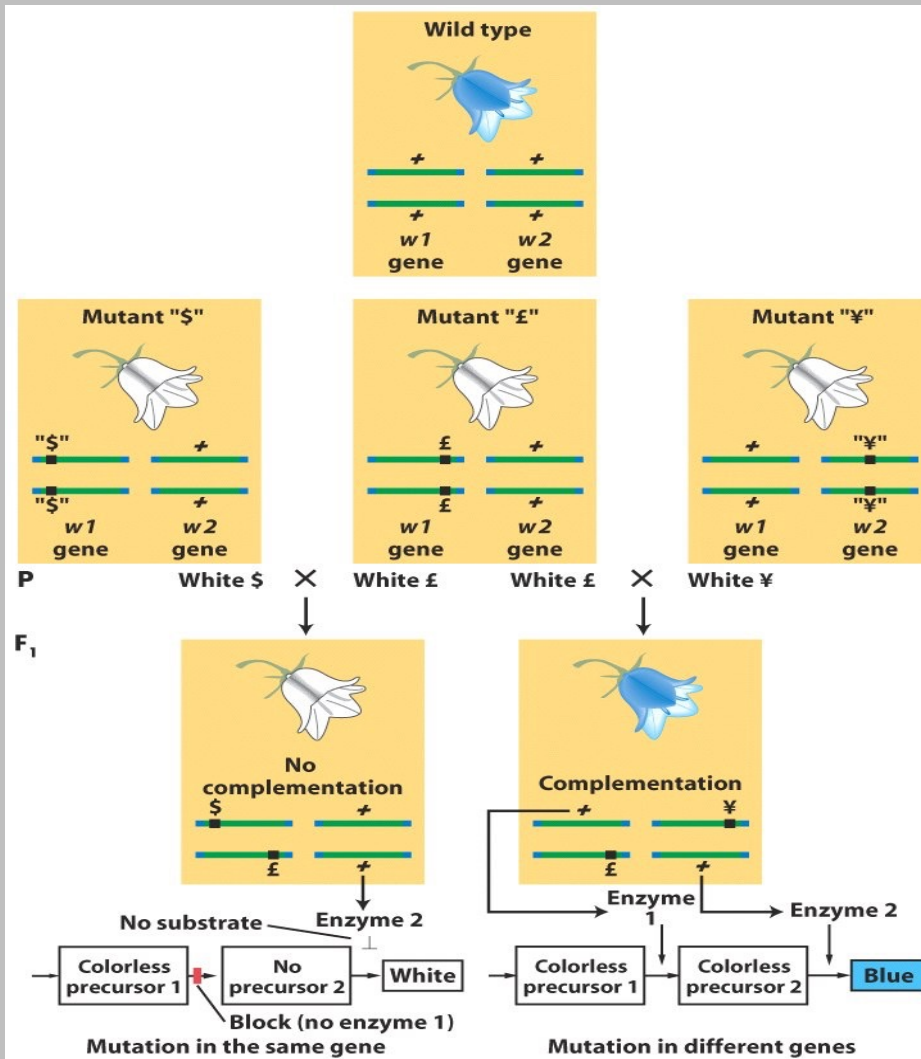
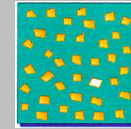
Beadle and Tatum

Um gene > uma enzima

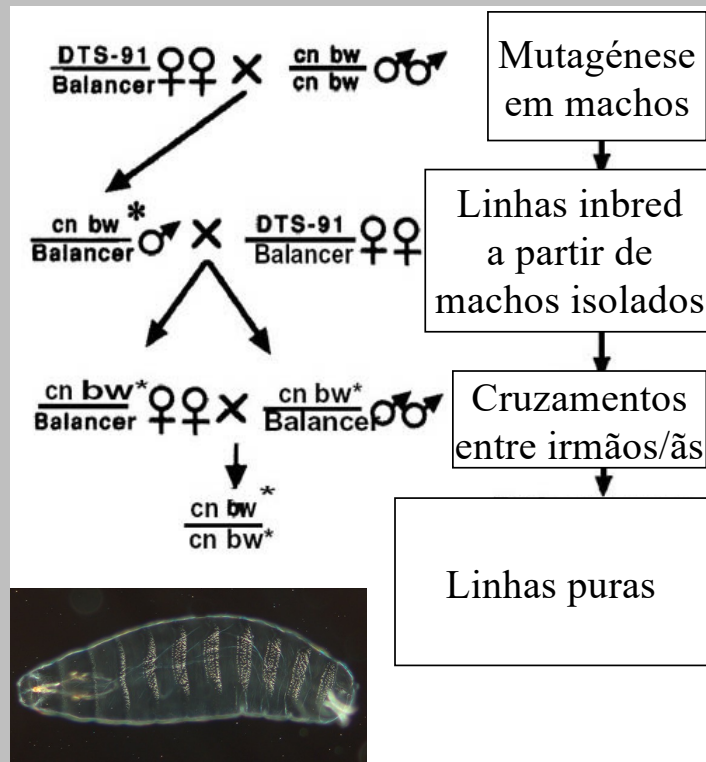
Um gene > uma cadeia polipeptídica



Complementação



Esquema de cruzamentos para identificação de mutantes no cromossoma II



Results of mutagenesis screens

Total lines established and tested	26978
Lethal mutations	18136
Mutations causing embryonic lethality	4332
Mutations causing embryonic phenotypes	580
Complementation Groups	139



Ed Lewis, Christiane Nüsslein-Volhard and Eric Wieschaus
Nobel Prize 1995



Phenotype = Genetics + Environment + GxE